FET Switch R2.0

General-Purpose High-Speed SPDT Electrode Switch



GAA Custom Electronics, LLC
Kennewick, WA • www.gaacustom.com

Document revised September 2026 — packaging update

Overview

The FET Switch R2.0 is a general-purpose, high-speed solid-state SPDT (single-pole, double-throw) switch designed to connect an electrode or other load to one of two signal sources under logic control. It is intended as a building block for instrument designers and researchers who need to route a signal — AC, DC, or a mix of both — onto a load, quickly and cleanly, without the contact bounce, wear, or size of a mechanical relay.

Because the switching element is a FET-based analog switch rather than a relay, the module can pass fast analog waveforms as well as static DC levels, and it can do so at switching rates and rise/fall times far beyond what any mechanical switch can achieve. This makes it useful well beyond its original application in mass spectrometry and ion mobility instrumentation — anywhere an electrode, probe, or transmission line needs to be switched between two sources under electronic control.

Typical Application

A common use case is switching an electrode between two very different signal types on the same node — for example, alternating a Twave/RF drive signal with a DC blocking voltage on the same electrode, as shown below. The COM terminal connects to the load (the electrode). The NO and NC terminals connect to the two sources being switched between. A 0–5V logic-level control signal at the switch's control input determines which source is connected to COM.

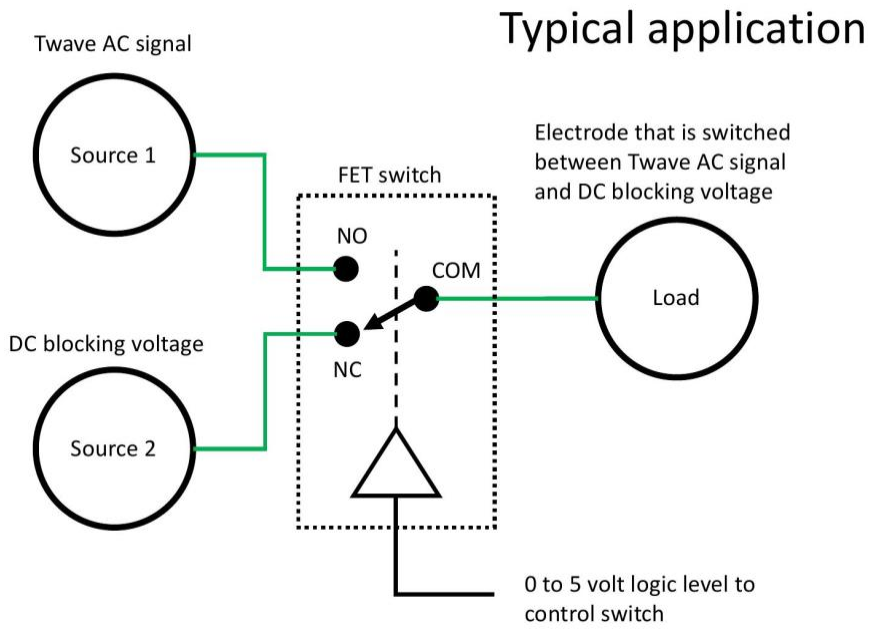


Figure 1 — Typical application: switching an electrode between two signal sources.

This is only one example. The same module is equally useful for routing test signals, isolating a sensor from a bias source, multiplexing analog inputs, or any other application requiring fast, floating, electronically-controlled signal switching.

Key Specifications

- High-speed FET SPDT switch
- Common (COM) output, Normally Closed (NC) input, Normally Open (NO) input
- Rise and fall times ~15 ns
- Passes analog signals as well as static DC levels
- Floatable to 3 kV

- Maximum voltage difference between the NO and NC signals: 300 V
- 0 to 5 volt logic-level control input, high-Z
- Powered from a 12 VDC supply
- Overvoltage protection using 300 V gas discharge tubes (GDTs)
- Capable of dissipating 5 W of power

Note: the electrical design, ratings, and performance above are unchanged from the original R2.0 release. The changes described in this document are limited to packaging, connectors, and available accessory options.

Power Dissipation

The switch is rated to dissipate up to 5 W. In most applications, the switch drives an electrode that presents a capacitive load, so the power dissipated depends on the load capacitance, the switching voltage, and the switching frequency rather than on a continuous current.

Power can be estimated from the following:

- Energy stored in a capacitor, in joules: $J = \frac{1}{2}CV^2$, where C is the load capacitance in farads and V is the voltage in volts.
- A joule per second is a watt.
- V is the voltage difference between the two switch inputs — for example, +100 V on NO and −200 V on NC gives V = 300 V.
- Each switching event (a charge or a discharge) dissipates energy; a 10 kHz square wave drive represents 20,000 charge/discharge events per second.
- Because power scales with the square of the voltage, doubling the switched voltage quadruples the dissipated power.

Switch Logic and Control

The switch is controlled by a single 0–5 V logic-level input. On the internal PCB, an onboard jumper (JP1) sets whether a logic 1 (5 V) or a logic 0 (0 V) connects COM to NO. In the packaged modules described in this document, this jumper is set at the factory to logic 1 = ON (the standard configuration) unless a different configuration is requested at time of order.

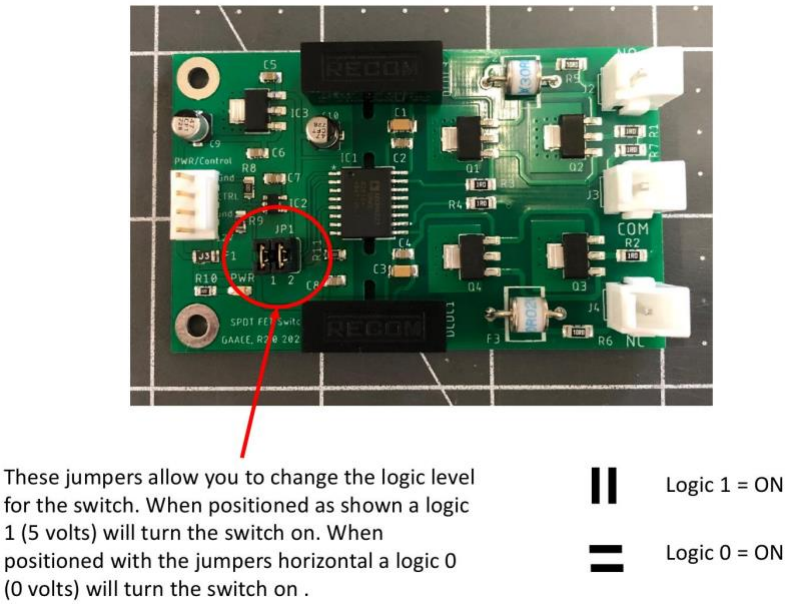


Figure 2 — Internal logic-select jumper (factory-set; shown for reference).

Packaging Options

The FET Switch R2.0 circuit is available in several packaged configurations. The two standard options are described below; GAA Custom Electronics can also build the switch into custom enclosures, connector styles, or cable sets to fit a customer's specific application — contact us to discuss requirements not covered by the standard options.

Single-Channel Module

The single-channel module packages one FET SPDT switch in a small enclosure with the GAA Custom Electronics logo. NO, COM, and NC signal connections are made through push-type connectors on one edge of the case; power and control are made through a 4-pin header on the opposite edge. A status LED bar indicates power and switch state.



Figure 3 — Single-channel module, standard enclosure.



Figure 4 — Single-channel modules as shipped, with accessory power/control cable and signal leads.

Single-Channel Power/Control Connector

Pin-out of the 4-pin power/control header:

Pin	Signal	Description
1	Ground	Logic and power ground
2	Control (0–5V, TTL)	Logic-level input; sets NO/NC connected to COM
3	Ground	Logic and power ground
4	+12V	+12 VDC power input

A wall-mount 12V power supply and a BNC-terminated control cable are provided with each module, along with a set of pigtail leads for the NO, COM, and NC connections. Keep the signal leads as short as practical to minimize ringing when switching, and insulate the module before applying power.

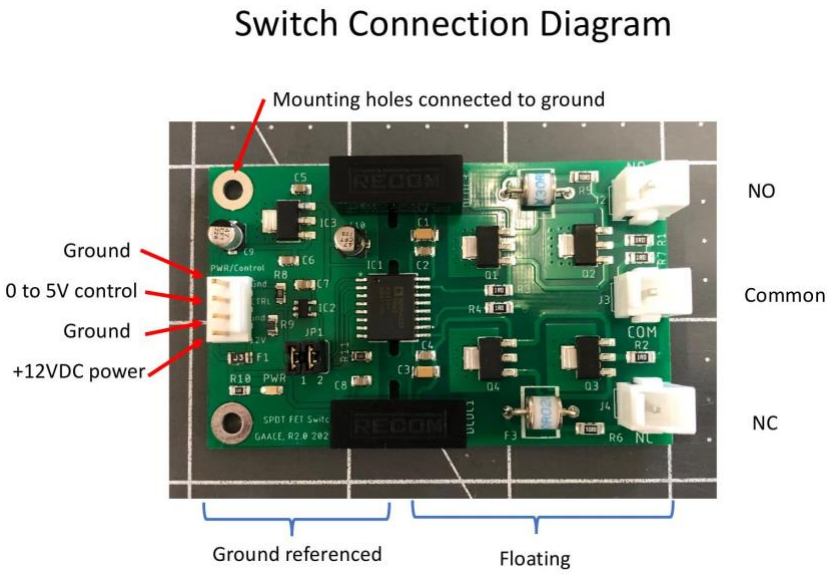


Figure 5 — Switch connections (internal PCB, shown for reference; connector locations are equivalent in the packaged module).

Dual-Channel Module (MIPS-Cabled, SMA)

The dual-channel module packages two independent FET SPDT switches (SW1 and SW2) in a single enclosure, intended for direct integration with a MIPS chassis. Each switch's NO, COM, and NC connections are brought out to its own SMA connector — six SMA connectors total. Power and the two control (trigger) signals are supplied through a single DB15 connector via a custom cable from the MIPS system, eliminating the need for a separate wall supply or BNC control cable.

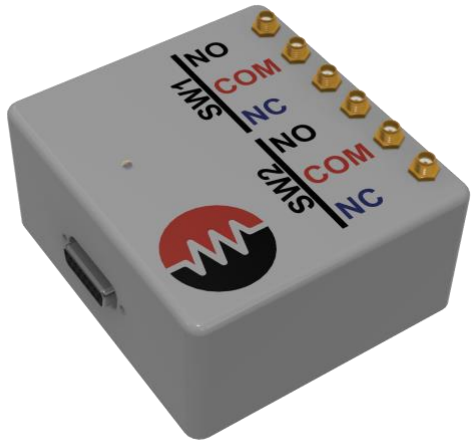


Figure 6 — Dual-channel module, SMA connectors with MIPS DB15 power/trigger cable.

MIPS DB15 Power/Trigger Connector Pin-out

The DB15 connector follows the standard MIPS accessory power/trigger pin-out. This two-channel module uses only pins 1 and 2 for control; the remaining trigger lines (pins 3–6) are passed through on the connector but are not used by this module.

Pin	Signal	Description
1	Trigger 1 (SW1 control)	0–5V control signal for switch SW1
2	Trigger 2 (SW2 control)	0–5V control signal for switch SW2
3 – 6	Trigger 3 – 6 (unused)	Additional MIPS trigger lines; not used by this module
7 – 8	+12V	+12 VDC power input (two pins)
9 – 13	Signal Ground	Logic/signal ground reference (five pins)
14 – 15	+12V Return	Power return / ground (two pins)

Other Configurations

The single-channel and dual-channel modules above cover the most common use cases, but the underlying FET switch circuit can be packaged with other connector types (SMA, BNC, push-type, terminal block), channel counts, cable/harness configurations, or enclosure styles. Contact GAA Custom Electronics to discuss a configuration for your application.

Schematic

The circuit is unchanged from the original R2.0 release across all packaging options described in this document.

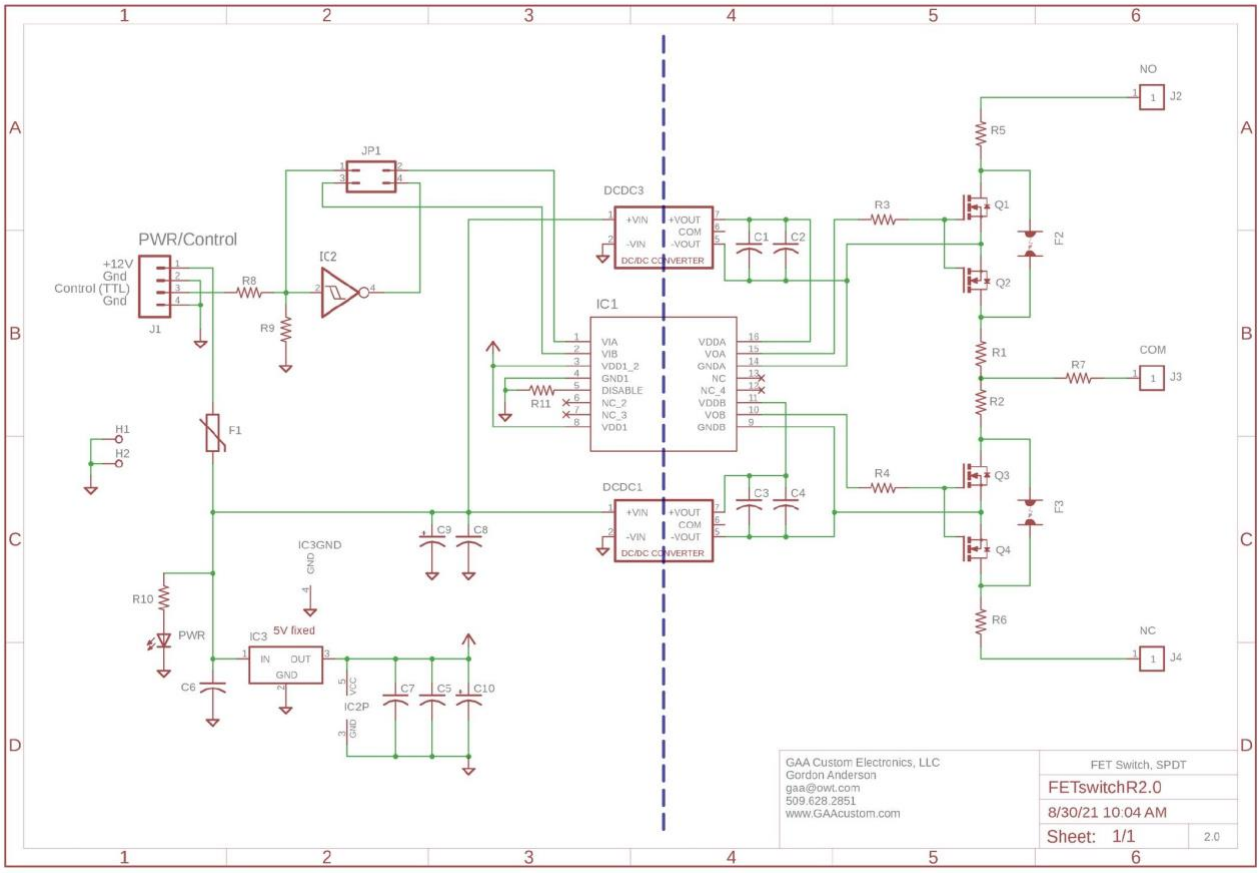


Figure 7 — FET Switch R2.0 schematic.

Bill of Materials

Component-level BOM for the FET Switch R2.0 PCB assembly, common to all packaging options.

Qty	Value	Device	Package	Ref.	Description	Part Number
2	10uF, 35V	C-USC1206	C1206	C1, C3	CAP CER 10UF 35V X6S 1206	490-12425-1-ND
6	0.1uF, 100V	C-USC0805	C0805	C2, C4, C5, C6, C7, C8	CAP CER 0.1UF 100V X7R 0805	1276-2447-2-ND
2	47uF, 16V	CPOL-USB	PANASONIC_B	C9, C10	CAP ALUM 47UF 20% 16V SMD	P15090CT-ND
2	12V in, 12V out	RP-XX	RP-XX	DCDC1, DCDC3	DC DC CONVERTER 12V 1W	945-1254-ND
1	—	SPARKFUN-POWERIC_PTCSMD	PTC-1206	F1	PTC RESET FUSE 30V 350MA 1206	F8473TR-ND
2	500V	ARRESTER	EPCOS_N81	F2, F3	GDT 300V 2.5KA 2 POLE TH	GTCA25-301M-R02-ND
1	—	ADUM4223CRWZ-RL	16-SOIC	IC1	DGTL ISO 5KV 2CH GATE DVR 16SOIC	ADUM4223CRWZ-RLCT-ND
1	—	NC7SZ14M5X	SOT23-5	IC2	TinyLogic UHS Inverter, Schmitt Trigger Input	NC7SZ14M5XCT-ND
1	5V	LT1129CST5	SOT223	IC3	Voltage regulator, 5V	296-12290-1-ND
1	—	M04POLAR	MOLEX-1X4	J1	Conn header 4POS .100 vert gold	WM2702-ND
3	—	1-1123824-1	TE_1-1123824-1	J2, J3, J4	Conn header, R/A, 1POS	A130985-ND
1	—	JP2Q	JP2Q	JP1	0.1" 2x40 pin strip	S2011EC-40-ND
1	—	LEDCHIP-LED0805	CHIP-LED0805	PWR	LED green diffused, 0805 SMD	475-1410-1-ND
4	—	N-HEXFET-2D	SOT223	Q1, Q2, Q3, Q4	MOSFET N-CH 450V 140mA SOT223	ZVN0545GCT-ND
5	1	R-US_R1206	R1206	R1, R2, R3, R4, R7	1 ohm, 1206	P1.0ECT-ND
1	1200	R-US_R0805	R0805	R10	1200 ohm, 0805	P1.2KACT-ND
1	1000	R-US_R0805	R0805	R11	1000 ohm, 0805	P1.0KACT-ND
2	10	R-US_R1206	R1206	R5, R6	10 ohm, 1206	P10.0FCT-ND
1	10	R-US_R0805	R0805	R8	10 ohm 5% 1/8W, 0805	P10ACT-ND
1	10K	R-US_R0805	R0805	R9	10K ohm 5% 1/8W, 0805	P10KACT-ND

Ordering and Custom Configurations

GAA Custom Electronics packages the FET Switch R2.0 to fit the customer's application — single- or multi-channel, with push-type, SMA, BNC, or terminal-block connectors, and with or without a MIPS-integrated power/trigger cable. Contact us to discuss a configuration not covered by the standard options in this document.

GAA Custom Electronics, LLC
Kennewick, WA

www.gaacustom.com